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The Rapid Adoption of Generative AI

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driven by differences in both the intensity of genAI use and labor cost shares. These factors can offset one another: for example, the information services sector has high rates of genAI use and time savings, but its estimated productivity gain is muted by its low labor cost share. At the aggregate level, we estimate a potential productivity gain of 1.2% based on current genAI use. This is close to Acemoglu (2024), who uses a similar framework to estimate a 0.7% gain based on predicted task exposure rather than actual adoption. Notably, however, Acemoglu (2024) assumed these productivity gains may take a decade to materialize. In contrast, our estimates reflect current adoption and time savings, implying a faster pace of diffusion. We also emphasize that our estimates do not capture potential productivity gains from using genAI to reorganize or completely automate production processes, and therefore likely represent a lower bound over longer horizons (Bresnahan and Trajtenberg, 1995; Brynjolfsson and Hitt, 2000; David, 1990; Jovanovic and Rousseau, 2005).

2 Data Sources and Measurement

2.1 The Real-Time Population Survey (RPS)

Our main data source is the Real-Time Population Survey (RPS), a national labor market survey of U.S. adults aged 18-64 (for a detailed discussion, see Bick and Blandin 2023). The RPS is fielded online by Qualtrics, a large commercial survey provider, and has collected multiple survey waves each year starting in 2020.

The RPS is designed to mirror the Current Population Survey (CPS) along key dimensions. The RPS matches questions on demographics and labor market outcomes in the basic CPS and CPS Outgoing Rotation Group, using the same word-for-word phrasing when practical and replicating the intricate sequence of questions necessary to elicit labor market outcomes in a manner consistent with the CPS (US Census Bureau, 2015). Replicating key portions of an existing high-quality survey ensures that survey concepts are comparable, which allows researchers to validate RPS outcomes against a widely used benchmark with a larger sample size and to construct sample weights. Bick and Blandin (2023) and Bick et al. (2025a,b) show that the RPS closely aligns with government household surveys on distributions of employment, hours worked, earnings, industry composition, employee tenure, and work from home.

In June 2024, the RPS introduced a novel module measuring genAI use both at work and at home.

2.1.1 RPS Sample

Qualtrics panel members are recruited to the panel online and, in our case, can participate in exchange for 30 to 50 percent of the \$5.50 paid per completed survey. The Qualtrics panel includes about 15 million members and is not a random sample of the U.S. population. However, Qualtrics can target survey invitations to specific demographic groups. The RPS sample was designed to be nationally representative across key demographics. We also include measures to filter out low quality responses. Details on sampling and data quality procedures can be found in Appendix [A.1](#).

We fielded the RPS with a pilot genAI module in June 2024 and received 2,551 responses. We then launched the full genAI module with the August and November 2024 waves and received 5,014 and 5,329 responses respectively. All surveys started fielding during the same weeks that the CPS conducted its corresponding surveys. Because the results are very similar across all surveys but our June pilot featured fewer questions, we report results from the pooled August and November surveys. Appendix [B](#) replicates the main figures of the paper separately by survey waves.

Appendix Table [A.1](#) compares the unweighted sample composition between the CPS and RPS along the demographics targeted in the sampling procedure for our main surveys (columns 1 and 2). The bottom panel also compares employment status in the CPS and RPS, which was not targeted in the sampling procedure. The two surveys yield very similar population shares.

2.1.2 Sample Weights and Validation

To address remaining discrepancies, we construct sample weights using the raking algorithm of Deming and Stephan ([1940](#)), ensuring that the weighted sample proportions align with the demographic characteristics targeted in the sampling procedure and that the employment shares in the RPS align with the CPS as well. Details on weighting can be found in Appendix [A.2](#).³

In Appendix [A.3](#), we show that the distribution of usual weekly earnings, occupation, and industry shares in the RPS, which were not targeted during sampling, closely align with the CPS, in both the raw and weighted data.

³In particular, Appendix [A.2.1](#) discusses a weighting adjustment we applied based on an experimental survey which randomized the sequencing of questions in the genAI module.

2.2 Measurement of Generative AI Use

When designing our survey, a key goal was to enable comparisons to the historical adoption of other technologies. To facilitate these comparisons, we use existing technology adoption questions from the CPS Computer and Internet Use supplement (CIU) as a template for our own questions regarding genAI. In the CIU, the leading question about computer adoption at work for employed respondents was:

Do you [directly] use a computer for your job? (No/Yes)

A second question asks about computer use at home:

Do you [directly] use a computer at home? (No/Yes)

These questions were asked in 1984, 1989, 1993, 1997, 2001, and 2003, with the word “directly” being omitted in the last two years.⁴

The RPS genAI module begins with a definition of genAI:

Generative AI is a type of artificial intelligence that creates text, images, audio, or video in response to prompts. Some examples of Generative AI include ChatGPT, Gemini, and Midjourney.

We included examples of popular genAI products because we thought some respondents may be more familiar with those product names than with the broader concept of genAI. After defining genAI, the module asks respondents whether they had heard of the concept prior to the survey. Overall, 88.4% reported having heard of genAI; the remaining 11.6% who answered “No” skip the remainder of the module. Respondents who answer “Yes” continue with the survey.

For employed respondents, the next question asks about genAI use at work:

Do you use Generative AI for your job? (No/Yes)

This question is designed to mirror the analogous computer use question from the CIU, discussed above. We opted for the version asked from 2001 onwards that omitted the word “directly.”

Another question later in the survey asks about genAI usage outside work:

Do you use Generative AI [outside your job]? (No/Yes)

⁴Starting in 2001, the CIU also asked about internet usage.

Non-employed respondents are not shown the term in brackets. While the CIU asks about computer use “at home”, we use slightly different language to account for respondents who use genAI on mobile devices outside of their home.

The survey asks several follow-up questions of genAI users regarding which products they used, how intensively they used it, and how much time they saved, always distinguishing between work and non-work use.

An important caveat regarding our measurement approach is that it will only capture genAI use that respondents are aware of. For example, while typing prompts into ChatGPT is an obvious case of using genAI, automatic AI-assisted writing suggestions are a more subtle case that respondents may not consider. Because our estimates may not detect some passive or embedded uses of the technology, we interpret our estimates a lower bound on the extent of genAI adoption.

3 How Prevalent is Generative AI Use?

Figure 1a displays the share of August and November 2024 RPS respondents who report using genAI. The samples for the “Overall” and “Outside of Work” bars are all respondents, while the sample for the “For Work” bar is employed respondents. The overall estimates reflect the share who use genAI either for work or outside of work.

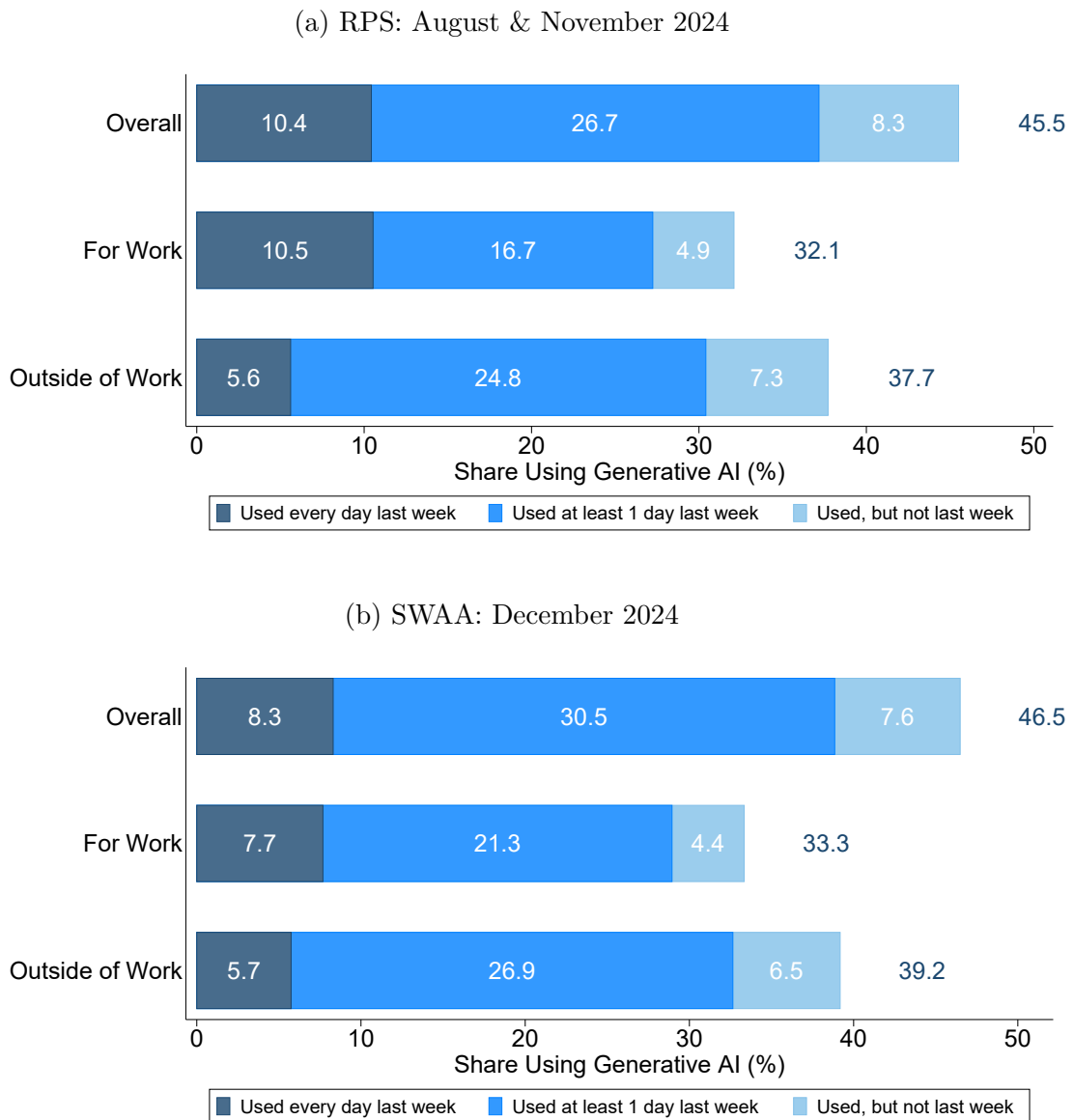
The first bar shows that 45.5% of respondents report using genAI either at work or at home: 10.4% used it every day, 26.7% used it some but not all days, and 8.3% did not use it in the last week. The second bar shows that 32.1% of workers used genAI at work: 10.5% used it every work day and 16.7% on some but not all workdays, bringing total weekly usage to 27.2%. Outside of work, usage is higher overall (37.7%) but less intensive, with only 5.6% of all respondents using it every day last week.

3.1 How Do Our Results Compare to Other Estimates?

The RPS is administered by Qualtrics. To provide a second source of data, in December 2024 we fielded our genAI module through the Survey of Working Arrangements and Attitudes (SWAA) by Barrero et al. (2021).⁵ The SWAA uses a different survey provider (IncQuery) but targets a similar sample intended to be representative of the U.S. population ages 20-64. Figure 1b displays analogous results from the SWAA. 46.5% of SWAA respondents reported using genAI overall, versus 45.5% of RPS respondents. We find similar estimates for work (33.3% SWAA,

⁵See Appendix C for more details on the SWAA, how we construct weights to maximize comparability between the SWAA and RPS, and some validation exercises.

Figure 1: Share of Working Age Adults Using Generative AI



Notes: The figure shows the share of respondents who use genAI for work, outside of work, and overall (either for work or outside of work). Intensity of use is broken down into every day last week (dark blue), at least one day but not every day last week (medium blue), and not last week (light blue). Data source for panel (a) is the August and November 2024 waves of the RPS, ages 18-64. The “For Work” sample is employed individuals ($N = 6935$); the other bars include all respondents ($N = 9726$). Data source for panel (b) is the December 2024 wave of the SWAA, ages 20-64. The “For Work” sample is employed individuals ($N = 3516$); the other bars include all respondents ($N = 4698$).

32.1% RPS) and outside work (39.2% SWAA, 37.7% RPS). Appendix C.3 includes additional comparisons between the SWAA and RPS. Because the two surveys yield similar estimates but the SWAA only includes a subset of the genAI questions in the RPS, the remainder of the main text reports results only from the RPS.